

Edexcel / Pearson GCSE Business — Year 11

Theme 1 & Theme 2 · Every formula that may appear on your exam paper

FORMULA NAME	FORMULA	NOTES / UNITS
■ 1 · Revenue, Costs & Profit		
Total Revenue (TR)	$TR = \text{Price (P)} \times \text{Quantity Sold (Q)}$	Answer in £
Total Costs (TC)	$TC = \text{Fixed Costs (FC)} + \text{Variable Costs (VC)}$	Answer in £
Total Variable Costs	$\text{Total VC} = \text{Variable Cost per Unit} \times \text{Output}$	Answer in £
Profit / Loss	$\text{Profit} = \text{Total Revenue} - \text{Total Costs}$	+ = profit – = loss
Gross Profit	$\text{Gross Profit} = \text{Revenue} - \text{Cost of Sales}$	Answer in £
Net Profit (Operating Profit)	$\text{Net Profit} = \text{Gross Profit} - \text{Other Operating Expenses}$	Answer in £
■ 2 · Profit Margins (%)		
Gross Profit Margin	$\text{GPM (\%)} = (\text{Gross Profit} \div \text{Revenue}) \times 100$	Express as %
Net Profit Margin	$\text{NPM (\%)} = (\text{Net Profit} \div \text{Revenue}) \times 100$	Express as %
■ 3 · Break-Even Analysis		
Contribution per Unit	$\text{Contribution} = \text{Selling Price per Unit} - \text{Variable Cost per Unit}$	Answer in £ per unit
Break-Even Output	$\text{BEO} = \text{Fixed Costs} \div \text{Contribution per Unit}$	Answer in units
Total Contribution	$\text{Total Contribution} = \text{Contribution per Unit} \times \text{Units Sold}$	Answer in £
Margin of Safety (units)	$\text{MoS} = \text{Actual Output} - \text{Break-Even Output}$	Answer in units
Margin of Safety (%)	$\text{MoS (\%)} = (\text{MoS units} \div \text{Actual Output}) \times 100$	Express as %
■ 4 · Cash Flow		
Net Cash Flow	$\text{Net Cash Flow} = \text{Cash Inflows} - \text{Cash Outflows}$	Answer in £
Closing Balance	$\text{Closing Balance} = \text{Opening Balance} + \text{Net Cash Flow}$	Answer in £
Opening Balance	$\text{Opening Balance} = \text{Previous Period's Closing Balance}$	Carry forward figure
■ 5 · Investment Appraisal — ARR		
Total Profit over Investment	$\text{Total Profit} = \text{Total Returns} - \text{Initial Investment}$	Answer in £

Average Annual Profit	Avg Annual Profit = Total Profit ÷ Number of Years	Answer in £ per year
Average Rate of Return (ARR)	ARR (%) = (Avg Annual Profit ÷ Initial Investment) × 100	Compare to interest rate

■ 6 · Market Calculations

Market Share	Market Share (%) = (Sales of Business ÷ Total Market Sales) × 100	Express as %
Market Growth	Market Growth (%) = ((New Size – Old Size) ÷ Old Size) × 100	Express as %
Price Elasticity of Demand	PED = % Change in Quantity Demanded ÷ % Change in Price	Ignore the minus sign
% Change (general)	% Change = ((New – Old) ÷ Old) × 100	Express as %

■ 7 · Financial Ratios

Return on Investment (ROI)	ROI (%) = (Net Profit ÷ Investment) × 100	Express as %
Labour Productivity	Productivity = Output ÷ Number of Workers	Units per worker
Labour Turnover	Labour Turnover (%) = (Leavers ÷ Avg Employees) × 100	Express as % (lower = better)
Absenteeism Rate	Absenteeism (%) = (Days Absent ÷ Total Working Days) × 100	Express as %

■ 8 · Pricing Calculations

Cost-Plus / Mark-Up Pricing	Selling Price = Cost of Production + Mark-Up	Mark-up in £ or %
Mark-Up Percentage	Mark-Up (%) = (Profit per Unit ÷ Cost per Unit) × 100	Express as %

★ EXAM TIPS: (1) Always show your working — method marks are awarded even if the final answer is wrong. (2) Include units (£, %, units) in every answer. (3) Read the question carefully — 'calculate' means a numerical answer; 'explain' means context too. (4) For % change questions, identify 'old' and 'new' values before substituting.